

Therefore, for a 500 stock portfolio there will be 125,250 unique parameters. Since there are 500 data points per day we need just over 1 year of data (250 trading days per year) just to calculate each parameter in the covariance matrix.

But now, the problem associated with estimating each parameter entry in the covariance matrix is further amplified because we are not solving for a deterministic set of equations. We are seeking to estimate the value of each parameter. A general rule of thumb is that there needs to be at least 20 observations for each parameter to have statistically meaningful results.

The number of data points required is then:

$$\begin{aligned}d &\geq 20 \cdot k \\n \cdot t &\geq 20 \cdot \frac{n \cdot (n + 1)}{2} \\t &\geq 10 \cdot (n + 1)\end{aligned}$$

Therefore, for a 500 stock portfolio (the size of the market index) we need 5010 days of observations, which is equivalent to over 20 years of data! Even if we require only 10 data points per parameter this still results in over 10 years of data! [Figure 6.8b](#) shows the number of days of data that is required to estimate the parameters of the covariance matrix for different numbers of stocks.

It has been suggested by some industry pundits that it is possible to estimate all unique parameters of the covariance matrix using the same number of observations as there are unique parameters. However, these pundits also state that in order for this methodology to be statistically correct we need to compute the covariance terms across the entire universe of stocks and not just for a subset of stocks. But even if this is true, the relationship across companies in the methodology needs to be stable. The reasoning is that if we do use the entire universe of stocks with enough data points we will uncover the true intrarelation across all subgroups of stocks and have accurate variance and covariance measures.

In the US there are over 7000 stocks and thus over 24.5 million parameters. This would require over 14 years of data history! We are pretty confident in the last 14 years that many companies have changed main lines of products (e.g., Apple), changed their corporate strategy (e.g., IBM), and thus these relationships have changed. So even if we had enough data points we know that companies do change, violating the requirements for this approach.